CHE 230: Molecular Systems in Chemical Engineering Homework 2

Due: 2/27/2026 23:59.

CHE 230

1. Of the metals listed in the Table below, (12 points)

   (a) Which will experience the greatest percent reduction in area? Why?
   B
   It has the highest strain at fracture (0.40), so it's the most ductile and necks the most before breaking.

   (b) Which is the strongest? Why?
   D
   It has the highest tensile strength (850 MPa) (and the highest yield strength), so it can carry the most stress.

   (c) Which is the stiffest? Why?
   E
   It has the highest elastic modulus (350 GPa), so it resists elastic stretching the most.

   (d) Which is the hardest? Why?
   D
   Hardness generally tracks with high strength (especially yield strength), and D has the highest yield strength (700 MPa).

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Material	Yield Strength (MPa)	Tensile Strength (MPa)	Strain at Fracture	Fracture Strength (MPa)	Elastic Modulus (GPa)
A	310	340	0.23	265	210
B	100	120	0.40	105	150
C	415	550	0.15	500	310
D	700	850	0.14	720	210
E	Fractures before yielding			650	350

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2. Consider a cylindrical titanium wire 3.0 mm (0.12 in.) in diameter and $2.5 \cdot 10^4$ mm (1000 in.) long. Calculate its elongation when a load of 500 N (112 lb_f) is applied. Assume that the deformation is totally elastic.(8 points)

$$\varepsilon = \frac{\Delta L}{L}, \quad \sigma = E\varepsilon, \quad \sigma = \frac{F}{A}$$

$L = 25$ m

$F = 500$ N

$A = \pi r^2 = \frac{\pi D^2}{4} = \frac{\pi}{4}(\cancel{3.0 \times 10^{-3}})^2$ m

Elastic Deformation for titanium:

$\approx E = 105 \times 10^9$ Pa $\left(\cancel{\frac{N}{m^2}}\right)$

$$\Rightarrow \frac{F}{A} = E \cdot \frac{\Delta L}{L}$$

$$\Rightarrow \frac{FL}{A \cdot E} = \Delta L$$

$$\Rightarrow \Delta L = \frac{500\,\cancel{N}\,(25\,m)}{\frac{\pi}{4}(0.003)^2\,\cancel{m^2} \cdot 105 \times 10^9\,\cancel{\frac{N}{m^2}}} = 0.0168\ m$$

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3. In chemical plants, metallic rods and wires are cold-drawn to improve strength for applications such as heat exchanger tubing, instrumentation lines, and structural supports. Three candidate materials are being considered:

a. Copper

b. Brass

c. 1040 Steel

Each material is drawn from an initial diameter of 12.0 mm to a final diameter of 10.4 mm.

   a. Calculate the Percent Cold Work (%CW) produced by the drawing process (2 points).
   b. Using the provided cold-work diagrams, estimate the following properties at the calculated %CW for:
      i. Copper

ii. Brass

iii. 1040 Steel

For each material, determine:

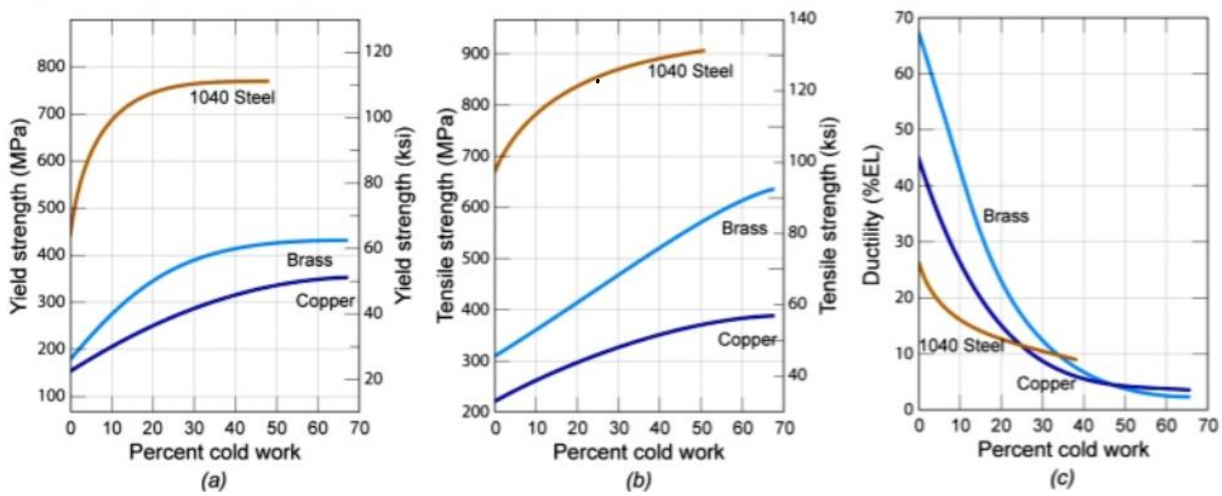
I. Yield Strength (MPa)

II. Tensile Strength (MPa)

III. Ductility (% Elongation) (9 points)

c. Based on your results, briefly state:

- Which material is most suitable for high-pressure service
- Which material is best where ductility/formability is important (Answer in 2–3 sentences.) (3 points)



a) $\% CW = \frac{A_o - A_f}{A_o}(100)$ A is proportional to d^2

$$\% CW = \frac{d_o^2 - d_f^2}{d_o^2} \times 100 = \frac{12^2 - 10.4^2}{12^2}(100) = 24.89\%$$

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b) @ ~25% CW

		Yield Strength (MPa)	Tensile Strength (MPa)	Ductility (% Elongation)
i.	Copper	270	320	13
ii.	Brass	370	450	20
iii.	1040 Steel	750	850	12

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c) - Best for high pressure service: 1040 steel because it has the highest yield & tensile strength @25% CW.
- Best where ductility matters: brass because it keeps the highest % elongation (is the most ductile)

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4. A single crystal of some metal has a τ_{crss} of 30.7 MPa and is exposed to a tensile stress of 61 MPa.

(i) Will yielding occur when ϕ = 56°and λ = 32°. (3 points)

(ii) If not, what stress is necessary? (3 points)

given $\tau_{CRSS} = 30.7$ MPa

$\sigma = 61$ MPa

Shear stress: $\tau_R = \sigma \cos(\phi°)\cos(\lambda°) = 61 \text{ MPa} \cos(56°)(\cos(32°) = 28.93$ MPa

i) $\tau_R < \tau_{CRSS}$ ⇒ yielding will not occur

ii) Stress needed to yield:

$$\sigma_y = \frac{\tau_{CRSS}}{\cos\phi \cos\lambda} = \frac{30.7 \text{ MPa}}{\cos(56°)\cos(32°)} = 64.74 \text{ MPa}$$

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5. a. How are dislocations involved in the plastic deformation of metals/metal alloys? (4 points)

- A dislocation is a small defect in a metal's crystal structure.
- Plastic deformation happens when dislocations move through the crystal.
- Their movement lets layers of atoms slide past each other permanently.
- This requires much less force than moving all atoms at once.
- Blocking dislocation motion (alloying, hardening, grain boundaries) makes metals stronger.

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b. Does the crystal structure of a metal affect its mechanical characteristics? If so, how and why (4 points)?

- Yes, crystal structure strongly affects mechanical properties.
- Different structures have different numbers of slip systems which affects how easily dislocations move.
- FCC metals (like aluminium, copper) have many slip systems which makes them ductile and easy to shape.
- BCC metals (like iron at room temp) have fewer active slip systems which makes them stronger but less ductile.

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6. a. . How are mechanical properties affected by dislocation mobilities?(4 points)

- Mechanical properties depend on how easily dislocations can move through the crystal. Obstacles to dislocation motion (alloying atoms, grain boundaries, hardening) increase hardness and yield strength.
- If dislocations move easily, the metal bends more easily leading to lower strength and greater ductility.
- If dislocations are hard to move, the metal resists deformation leading to higher strength, lower ductility.

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b. What techniques are used to increase the strength/hardness of metals/alloys?(4 points)

- Strain hardening (cold working): Plastic deformation at room temperature increases dislocation density. The dislocations tangle and become harder to move, so the metal becomes stronger and harder.
- Solid solution hardening: Adding other atoms such as carbon in iron distorts the crystal lattice. This makes dislocation motion more difficult and increases strength.
- Grain size hardening: Smaller grains create more grain boundaries. Grain boundaries block dislocations and make the metal stronger.
- Annealing: Heating reduces defect and dislocation density and lowers strain energy. The metal becomes softer and more ductile.

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c. How are mechanical characteristics of deformed metal specimens altered by heat treatments? (4 points).

- Heat treatment usually reverses strain hardening by reducing defects and making dislocations easier to move.

  Some examples:
- Annealing: Heating reduces defect and dislocation density and lowers strain energy. The metal becomes softer and more ductile.
- Recovery: Some internal stresses are removed with little change in grain structure. Strength drops slightly, ductility increases.
- Recrystallization: New strain-free grains form and replace deformed grains. Strength and hardness decrease, ductility increases.
- Grain growth (if heated for longer): Grains get larger, which further reduces strength and hardness but increases ductility.

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